

Validation outcomes preserve the difference between nomination and confirmation

	Retrospective nomination	Additional model evidence	Frozen primary test	Human disease evidence	Independent replication
21-gene network panel	Nominated 21 fixed genes	Not confirmed 0 FDR genes in GSE46798/GSE17542	Not tested	Not confirmed 0 meta-FDR genes	Not tested
Continuous network score	Nominated Minimum-distance proximity	Mixed / weak max partial r = 0.031	Not tested	Not confirmed partial r = 0.010	Not tested
Proteasome assembly	Retrospective signal meta-FDR = 0.0020	Mixed / weak GSE150005 rotenone p = 0.240 day-8 nominal support	Not confirmed GSE116280 p = 0.0958	Confirmed GSE178265 p = 0.0352	Not confirmed GSE243639 p = 0.5095
Cilium / Hedgehog / trafficking	Nominated GSE150005 pathway family	Not tested	Not confirmed GSE196190 p = 0.1221	Not tested	Not confirmed LUHMES p = 0.392 to 0.401

Outcomes are assigned from the frozen plans and tracked result tables; null tests are retained.

Nominated / retrospective signal Confirmed Mixed / weak Not confirmed Not tested